



PhysicalWorks Forge 주요기능 Manual

1. Data Upload
2. Data Collector
3. Data Generator
4. Data Manager
5. Model Trainer
6. Model Simulation
7. Model Manager

0. Project 생성/조회/변경 기능

최초 RFM 개발 프로젝트를 등록, 로그인 시 등록된 Project의 목록을 확인하고, 선택 및 변경하는 기능

- 1) Project 명을 선택 > 등록되어 있는 Project 목록 확인 가능
- 2) 현재 Project 이외에 다른 Project를 선택 시, 해당 Project로 이동

The screenshot displays the PhysicalWorks Forge web application. A modal window titled "Project" is open, showing a list of projects. The "Vega Logistics Center Project" is highlighted with a red box and a red circle with the number 2. The modal also shows a "G1 Logistics Center Project". The background interface includes a sidebar with navigation options like Dashboard, Data Foundry, Data Upload, Data Collector, Data Generator, Data Curator, Data Manager, Model Institute, Model Trainer, Model Evaluator, Model Simulation, and Model Manager. The main content area shows a progress bar for "Robot Learning & Training Status" with steps 03 to 06, a "Supported Foundation Models" section with GROOT N1.5, Open VLA, and π0.5, a "Finetuned Models" section with 147 models (+12% from last month), a "Dataset" section with a line graph showing an upward trend, and a "To-Do List" section with 4 In Progress, 1 Delayed, and 87.5% Completed tasks.

PhysicalWorks Forge

Project

Projects 2

- Vega Logistics Center Project**
Humanoid Dexmate Vega
Dexmate Vega 수화물 적재 학습 프로젝트
- G1 Logistics Center Project**
Humanoid Unitree G1
Unitree G1 수화물 적재 학습 프로젝트

Supported Foundation Models

- GROOT N1.5
- Open VLA
- π0.5

Finetuned Models

147
+12% from last month

Dataset

Line graph showing dataset growth over the week (Mon to Sun). The y-axis ranges from 0 to 70. The data points are approximately: Mon (8), Tue (22), Wed (22), Thu (32), Fri (38), Sat (48), Sun (62). The graph shows a +12% increase from last month. Total 167, Augmented 89.

To-Do List

- In Progress: 4
- Delayed: 1
- Completed: 87.5%

Navigation: Model v4

1. Data Upload

기존에 수집된 학습데이터 또는 Data Collector(Teleop)로 수집한 학습데이터를 플랫폼에 업로드하는 기능

1) Data Upload 메뉴 선택

2) 기본 정보(Basic Settings) 입력 > 업로드 할 파일 선택(Zip형태 또는 Directory) > Register 버튼 선택

The screenshot displays the PhysicalWorks Forge interface for data upload. The sidebar on the left contains the 'Data Foundry' menu, which is highlighted with a red box and a circled '1'. The main content area is titled 'Data Upload' and includes a subtitle 'Register new datasets with metadata, tags, and file upload.' The interface is divided into two steps: 'Step 1 Basic Settings' and 'Step 2 File Upload'. Step 1 is the active step, showing a form with the following fields: 'Dataset Name' (with example 'e.g., Manipulation Tasks v3'), 'Task' (with example 'e.g., Pick and Place'), 'Version' (with example 'e.g., 1.0.0'), 'Tags' (with placeholder 'Enter tag and press Enter'), 'Robot Model' (with a dropdown menu), 'Data Format' (with a dropdown menu), and 'Description' (with placeholder 'Enter description'). The entire form area is enclosed in a red box with a circled '2'. Step 2, 'File Upload', is partially visible below Step 1, showing a file selection area with a red box and a circled '3'. At the bottom right of the interface, there is a 'Register' button with a checkmark icon, highlighted with a red box and a circled '4'.

2. Data Collector (1/4)

Teleoperation 기능으로 로봇 동작 데이터를 수집하는 기능

1) Data Collector 메뉴 선택

2) Connect 버튼 선택 > 수집할 Episode 수 입력 > Record 버튼 선택

The screenshot shows the PhysicalWorks Forge Data Collector interface. The sidebar on the left has a 'Data Collector' menu item highlighted with a red box and a circled '1'. The main area is titled 'Data Collector' and shows a 'Record' step. A 'Connect' button is highlighted with a red box and a circled '2'. Below it, a 'Record' button is highlighted with a red box and a circled '4'. To the right of the 'Record' button, a 'NUM EPISODES' input field is highlighted with a red box and a circled '3', with the value '예: 10' entered. Below the input field, a 'Record' button is highlighted with a red box and a circled '4'. Below the 'Record' button, there are buttons for 'Stop', 'Retry', and 'Finish Session'. A red error message is displayed: 'Failed to connect to rosbridge. Check the URL and rosbridge server state.' At the bottom, there are three camera feeds: 'Head Camera', 'Left Wrist Camera', and 'Right Wrist Camera'. Each feed shows a camera icon and the text 'ROS bridge not connected'.

PhysicalWorks Forge

Vega Logistics Center Project > Humanoid Dexmate Vega

Dashboard

Data Foundry

Data Upload

Data Collector ①

Data Generator

- WFM Synthetic Data
- WFM Augmentation
- Mimic Augmentation
- Video to Motion

Data Curator

Data Manager

Model Institute

- Model Trainer
- Model Evaluator
- Model Simulation
- Model Manager

Data Collector

Record ROS episodes and review preview data for dataset collection.

Connection Error

1 Record

Live ROS ROS Stream Monitor

'Connect' 버튼으로 rosbridge에 연결하면 실시간 카메라 스트림과 주요 ROS 토픽 상태를 바로 확인할 수 있습니다.

② Connect Disconnect

Recorder Idle 0/0

Num Episodes Not set

Saved Episodes 0

Bag Root Loading...

NUM EPISODES

③ 예: 10

④ Record

Stop

Retry

Finish Session

'Record'는 episode 녹화를 시작하고, 'Stop'은 저장, 'Retry'는 현재 녹화본을 폐기합니다. 'Finish Session'은 목표 수량을 모두 채우지 않아도 현재까지 저장한 episode로 마감합니다.

Failed to connect to rosbridge. Check the URL and rosbridge server state.

Head Camera

/zed/zed_node/left/image_rect_color/compressed

Offline

ROS bridge not connected

토픽이 실제 publish되면 이 영역에 프레임이 표시됩니다.

Left Wrist Camera

/camera_left/camera_left/color/image_rect_raw/compressed

Offline

ROS bridge not connected

토픽이 실제 publish되면 이 영역에 프레임이 표시됩니다.

Right Wrist Camera

/camera_right/camera_right/color/image_rect_raw/compressed

Offline

ROS bridge not connected

토픽이 실제 publish되면 이 영역에 프레임이 표시됩니다.

2. Data Collector (2/4)

Teleoperation 기능으로 로봇 동작 데이터를 수집하는 기능

3) Episode 녹화 완료 > Stop 버튼 선택 > Episode 개수 만큼 반복 수행

4) 설정한 Episode 개수 이전에 종료를 원할 경우 > Finish Session 선택

The screenshot displays the PhysicalWorks Forge Data Collector interface. The left sidebar shows the navigation menu with 'Data Collector' selected. The main area is titled 'Data Collector' and includes a subtitle 'Record ROS episodes and review preview data for dataset collection.' A 'Connection Error' status is shown in the top right.

The interface is divided into two main sections: a left sidebar with steps and a main content area.

Left Sidebar (Steps):

- Step 1: Record (Active)
- Step 2: Dataset
- Step 3: Upload

Main Content Area (Record Step):

The main content area is titled 'Record' and features a 'ROS Stream Monitor' section. This section includes a 'Connect' button and a 'Disconnect' button. Below these, there are three status boxes:

- Recorder Idle: 0/0
- Num Episodes: Not set
- Saved Episodes: 0
- Bag Root: Loading...

On the right side of the ROS Stream Monitor, there are two buttons highlighted with red boxes and numbered:

- ⑤ Stop
- ⑥ Finish Session

Below the ROS Stream Monitor, there is a red error message: 'Failed to connect to rosbridge. Check the URL and rosbridge server state.'

At the bottom, there are three camera preview windows:

- Head Camera: /zed/zed_node/left/image_rect_color/compressed. Status: Offline.
- Left Wrist Camera: /camera_left/camera_left/color/image_rect_raw/compressed. Status: Offline.
- Right Wrist Camera: /camera_right/camera_right/color/image_rect_raw/compressed. Status: Offline.

Each camera preview window displays a camera icon and the message: 'ROS bridge not connected. 토픽이 실제 publish되면 이 영역에 프레임이 표시됩니다.'

2. Data Collector (3/4)

Teleoperation 기능으로 로봇 동작 데이터를 수집하는 기능

5) 조회할 에피소드를 선택 > Review 화면을 통하여 조회 및 검토

6) Convert to LeRobot 2.1 Dataset 버튼 선택

The screenshot displays the PhysicalWorks Forge Data Collector interface. The left sidebar shows the navigation menu with 'Data Collector' selected. The main area is divided into two steps: 'Record' and 'Dataset'. The 'Dataset' step is active, showing a 'Recorded Dataset Review' section with a table of datasets. The dataset '20260512_182713_episodes_003' is highlighted with a red box and a red circle labeled '7'. Below this, the 'Playback Review' section is shown with a red box and a red circle labeled '8'. It includes a timeline and three camera views: 'Head Camera', 'Left Wrist Camera', and 'Right Wrist Camera'. On the right side, the 'Convert to LeRobot 2.1 Dataset' button is highlighted with a red box and a red circle labeled '9'. The interface also shows a 'Dataset Catalog' section with a list of datasets and their completion status.

PhysicalWorks Forge

G1 Logistics Center Project

Humanoid Unitree G1

Dashboard

Data Foundry

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 - Model Manager

Data Collector 44 Record ROS episodes and review preview data for dataset collection.

Connection Error

Step 1 Record

Step 2 Dataset

Step 3 Upload

Dataset

Recorded Dataset Review

bag 폴더에 저장된 dataset을 선택하고 episode 단위로 검토할 수 있습니다.

Dataset Catalog

bag 폴더에 저장된 dataset을 선택합니다.

Dataset Name	Episodes	Status
20260513_151921_episodes_003	1/3 episodes	Complete
20260512_182713_episodes_003	3/3 episodes	Complete
20260512_182402_episodes_003	3/3 episodes	Complete

7

Playback Review

선택한 dataset의 녹화 프리뷰와 numeric signal을 검토합니다.

Episode 1 Play Reset

0.00s 4.62s 17.05s

Head Camera

Left Wrist Camera

Right Wrist Camera

Playback Metrics

검토할 numeric signal을 선택합니다.

pose_x /odom

pose_x -0.913 pose_y -0.203 linear_x 0.000 linear_y -0.000

Dataset Selected

Bag Root
/home/luke/PhysicalWorks_Forge/physicalw
orks-forge-recorder-local/bag/20260512_1
82713_episodes_003

Episodes
3/3

LeRobot 2.1 Dataset

Convert to LeRobot 2.1 Dataset

9

Playback Cameras

검토할 카메라 프리뷰를 선택합니다.

Head Camera
/zed/zed_node/left/image_rect_color/compressed

Left Wrist Camera
/camera_left/camera_left/color/image_rect_raw/compressed

Right Wrist Camera
/camera_right/camera_right/color/image_rect_raw/compressed

Playback Metrics

검토할 numeric signal을 선택합니다.

pose_x /odom

2. Data Collector (4/4)

Teleoperation 기능으로 로봇 동작 데이터를 수집하는 기능

7) 최종 데이터셋 정보를 확인 후, Upload 버튼 선택

The screenshot displays the PhysicalWorks Forge Data Collector interface. On the left is a dark sidebar with navigation options: G1 Logistics Center Project, Humanoid, Unitree G1, Dashboard, Data Foundry, Data Upload, Data Collector, Data Generator, Data Curator, Data Manager, Model Institute, Model Trainer, Model Evaluator, Model Simulation, and Model Manager. The main area is titled 'Data Collector' and shows a progress bar with three steps: Step 1 Record, Step 2 Dataset, and Step 3 Upload (the current step). The 'Upload' section contains a 'LeRobot Dataset List' with three entries, each showing episode and frame counts and a 'Ready' status. A red circle with the number 7 highlights the 'Upload' button in the bottom right corner. The interface also includes a 'Connection Error' message in the top right and a 'LeRobot Path' field.

PhysicalWorks Forge

G1 Logistics Center Project >
Humanoid Unitree G1

Dashboard

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WFM Synthetic Data

WFM Augmentation

Mimic Augmentation

Video to Motion

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Data Collector 44 Record ROS episodes and review preview data for dataset collection.

Connection Error

Step 1 Record

Step 2 Dataset

Step 3 Upload

Upload

LeRobot v2.1 형태로 변환된 dataset을 선택해서 dataset으로 업로드합니다.

LeRobot Dataset List

업로드 가능한 LeRobot dataset이 위에서 아래로 표시됩니다.

20260519_134933_episodes_001
/home/luke/PhysicalWorks_Forge/physicalworks-forge-recorder-local/bag/20260519_134933_episodes_001/lerobot_v21
1/1 episodes 196 frames Ready

20260519_105838_episodes_001
/home/luke/PhysicalWorks_Forge/physicalworks-forge-recorder-local/bag/20260519_105838_episodes_001/lerobot_v21
1/1 episodes 297 frames Ready

20260512_093222_episodes_003
/home/luke/PhysicalWorks_Forge/physicalworks-forge-recorder-local/bag/20260512_093222_episodes_003/lerobot_v21
3/3 episodes 312 frames Ready

Upload Ready

Selected Dataset
20260519_134933_episodes_001

LeRobot Path
/home/luke/PhysicalWorks_Forge/physicalworks-forge-recorder-local/bag/20260519_134933_episodes_001/lerobot_v21

REPO ID
LukeRobotics/INEW

HF TOKEN
hf_xxx

☐ Private dataset으로 업로드

→ Upload

Upload Status
ready
Repo LukeRobotics/INEW
upload completed
<https://huggingface.co/datasets/LukeRobotics/INEW>

3. Data Generator(#1) – WFM Synthetic Data(1/2)

Image 파일을 기반으로 (합성 데이터) 비디오 파일을 생성해 주는 기능

1) WFM Synthetic Data 메뉴 선택

2) 기본 정보 입력

3) Model Group 선택(Cosmos Predict 2.5) > Model 선택(Cosmos-Predict2.5-2B/Pre-trained 등...)

PhysicalWorks Forge

Vega Logistics Center Project >
Humanoid Dexmate Vega

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Model Manager

WFM Synthetic Data

Generate synthetic training data using world foundation models or augment existing demonstrations.

Step 1 Basic Settings

Step 2 Input Configuration

Step 3 File Upload

Step 4 Parameter Settings

1 Basic Settings

Dataset Name*
Enter dataset name

Task*
Select Task

Description
Enter description

2 Input Configuration

Model Group*
Select Model Group

Model*
Please select Model Group first

Model Type

H200 8 Total 3 Running 5 Idle
80GB HBM3e 4.8 TB/s 1,979 TFLOPS FP8
GPUs 1 2 3 4 5 6 7 8

H100 8 Total 3 Running 5 Idle
80GB HBM3 3.35 TB/s 1,979 TFLOPS FP8
GPUs 1 2 3 4 5 6 7 8

A100 8 Total 2 Running 6 Idle
80GB HBM2e 2.0 TB/s 624 TFLOPS FP16
GPUs 1 2 3 4 5 6 7 8

3 File Upload

Upload Image or Video*

Reset Generate

3. Data Generator(#1) – WFM Synthetic Data(2/2)

Image 파일을 기반으로 (합성 데이터) 비디오 파일을 생성해 주는 기능

4) 비디오 파일로 변환 할 Image 파일 업로드

5) 비디오 생성을 위한 Prompt 입력

6) Model Group 선택(Cosmos Predict 2.5) > Model 선택(Cosmos-Predict2.5-2B/Pre-trained 등...)

PhysicalWorks Forge

Vega Logistics Center Project

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WFM Synthetic Data

Generate synthetic training data using world foundation models or augment existing demonstrations.

Step 1
Basic Settings

Step 2
Input Configuration

Step 3
File Upload

Step 4
Parameter Settings

3 File Upload

Upload Image or Video *

4

Drag and Drop, or select from your local computer.

File Upload

Text Prompts (1/5) *

5

Prompt 1

Enter text prompt...

4 Parameter Settings

6

Negative Prompt

The video captures a series of frames showing ugly scenes, static with no motion, motion blur, over-saturation, shaky footage, low resolution, grainy texture, pixelated images, poorly lit areas, underexposed and overexposed scenes, poor color balance, washed out colors, choppy sequences, jerky movements, low frame rate, artifacting, color banding, unnatural transitions, outdated special effects, fake elements, unconvincing visuals, poorly edited content, jump cuts, visual noise, and flickering. Overall, the video is of poor quality.

Seed (0-2,147,483,647)

0

Guidance (0-7) *

7

Number of Steps (1-500) *

35

Resolution *

704,1280

Reset

7 Generate

3. Data Generator(#2) – Mimic Augmentation

등록된 시뮬레이션 데이터셋을 기반으로 대량의 데이터셋으로 증강 시켜주는 기능

1) Mimic Augmentation 메뉴 선택

2) 기본 정보 입력 > Task 선택 > 증강 할 데이터셋 선택

3) 시뮬레이션 환경 개수와 생성 할 데이터셋 개수 설정 > 실행 버튼 선택

PhysicalWorks Forge

Vega Logistics Center Project >
Humanoid Dexmate Vega

Dashboard

Data Foundry

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Mimic Augmentation

“ Augment existing demonstrations with variations in perspective, timing, and execution.”

Step 1 Basic Settings

Name *
Enter name

Description
Enter description

Step 2 Input Configuration

Task *
Select task

Datasets *
Please select task first
Please select dataset

Step 3 Parameter Settings

Number of Environments
1

Generation Num Trials *
10

☐ Enable Pinocchio IK ☐ Use SkillGen Motion Planner

Reset Execute

3. Data Generator(#3) – Video to Motion

(합성 데이터) 비디오 파일에 동작 궤적 데이터를 생성해주는 기능

1) Video to Motion 메뉴 선택

2) 기본 정보 입력

3) Model 선택

The screenshot displays the PhysicalWorks Forge interface for the 'Vega Logistics Center Project'. The left sidebar contains a navigation menu with the following items: Dashboard, Data Foundry, Data Upload, Data Collector, Data Generator (highlighted with a red box and circled 1), Data Curator, Data Manager, Model Institute, Model Trainer, Model Evaluator, Model Simulation, and Model Manager. Under 'Data Generator', 'Video to Motion' is selected and highlighted with a red box and circled 1.

The main panel is titled 'IDM Action Extraction' with the subtitle 'Extract actions from video datasets using Inverse Dynamics Model.' It features a three-step workflow on the left: Step 1 Basic Settings, Step 2 Input Configuration, and Step 3 Dataset Selection.

The 'Basic Settings' step (circled 2) includes a 'Name' input field, a 'Task' dropdown menu, and a 'Description' input field. The 'Input Configuration' step (circled 3) includes a 'Model' dropdown menu (circled 3) and a 'GPU Type' section with three options: H200, H100, and A100. Each GPU option shows its specifications (e.g., 80GB HBM3e, 4.8 TB/s, 1,979 TFLOPS FP8 for H200) and a status bar (8 Total, 3 Running, 5 Idle for H200). The 'Dataset Selection' step (circled 4) includes a 'Video Dataset' dropdown menu (circled 4).

At the bottom right, there are 'Reset' and 'Execute' buttons. The 'Execute' button is highlighted with a red box and circled 5.

4. Data Manager (1/4)

저장소에 등록되어 있는 데이터셋 목록을 조회하는 기능

1) Data Manager 메뉴 선택

2) Data 목록 중, 상세 정보 확인이 필요한 항목을 선택

PhysicalWorks Forge

Vega Logistics Center Project >
Humanoid Dexmate Vega

Dashboard

Data Foundry

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- Data Manager** ①
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- Model Manager

Data Manager “ Central repository for discovering, managing, and sharing robot learning datasets.

Q Search | Status All ▾

No.	Dataset Name	Description	Owner	Version	Records	Pre-Process	Source	Last Updated
1	G1 exhibition data v2.1	G1 exhibition data v2.1	developer	1.0.0	70	N	Register	2026-05-19 16:05:52
2	H1	H1 bottle picking sim data	developer	1.1.0	84	N	Validate	2026-05-12 16:44:40

Total 2 20 ▾ per page Go to Page

4. Data Manager (2/4)

저장소에 등록되어 있는 데이터셋 목록을 조회하는 기능

3) Episode Data 탭 선택

4) 영상과 동작 궤적 데이터를 확인하기 위한 Episode 항목을 선택

The screenshot displays the PhysicalWorks Forge Data Manager interface. On the left sidebar, the 'Vega Logistics Center Project' is selected, with 'Humanoid' and 'Dexmate Vega' sub-projects listed. The 'Data Foundry' section is active, showing a list of data sources including 'WFM Synthetic Data', 'WFM Augmentation', 'Mimic Augmentation', 'Video to Motion', 'Data Curator', 'Data Manager', 'Model Institute', 'Model Trainer', 'Model Evaluator', 'Model Simulation', and 'Model Manager'. The main panel shows the 'G1 exhibition data v2.1' dataset. The 'Episode Data' tab is selected and highlighted with a red box and a circled '3'. Below the tab, a table lists 70 episodes. The first five episodes (Index 0-4) are highlighted with a red box and a circled '4'. Each episode entry includes an index, a video thumbnail, a content description ('pick up cube.'), a status, and a duration.

PhysicalWorks Forge

Vega Logistics Center Project >

Humanoid Dexmate Vega

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- Model Manager

G1 exhibition data v2.1 G1 exhibition data v2.1

Edit Download

Overview Pipelines **Episode Data** Validation Result

G1 exhibition data v2.1

Owner: developer Version: 1.0.0 Records: 70 Tags: G1 LeRobot Teleoperation

Episode Data

Search 1.0.0

Showing 25 of 70 episodes

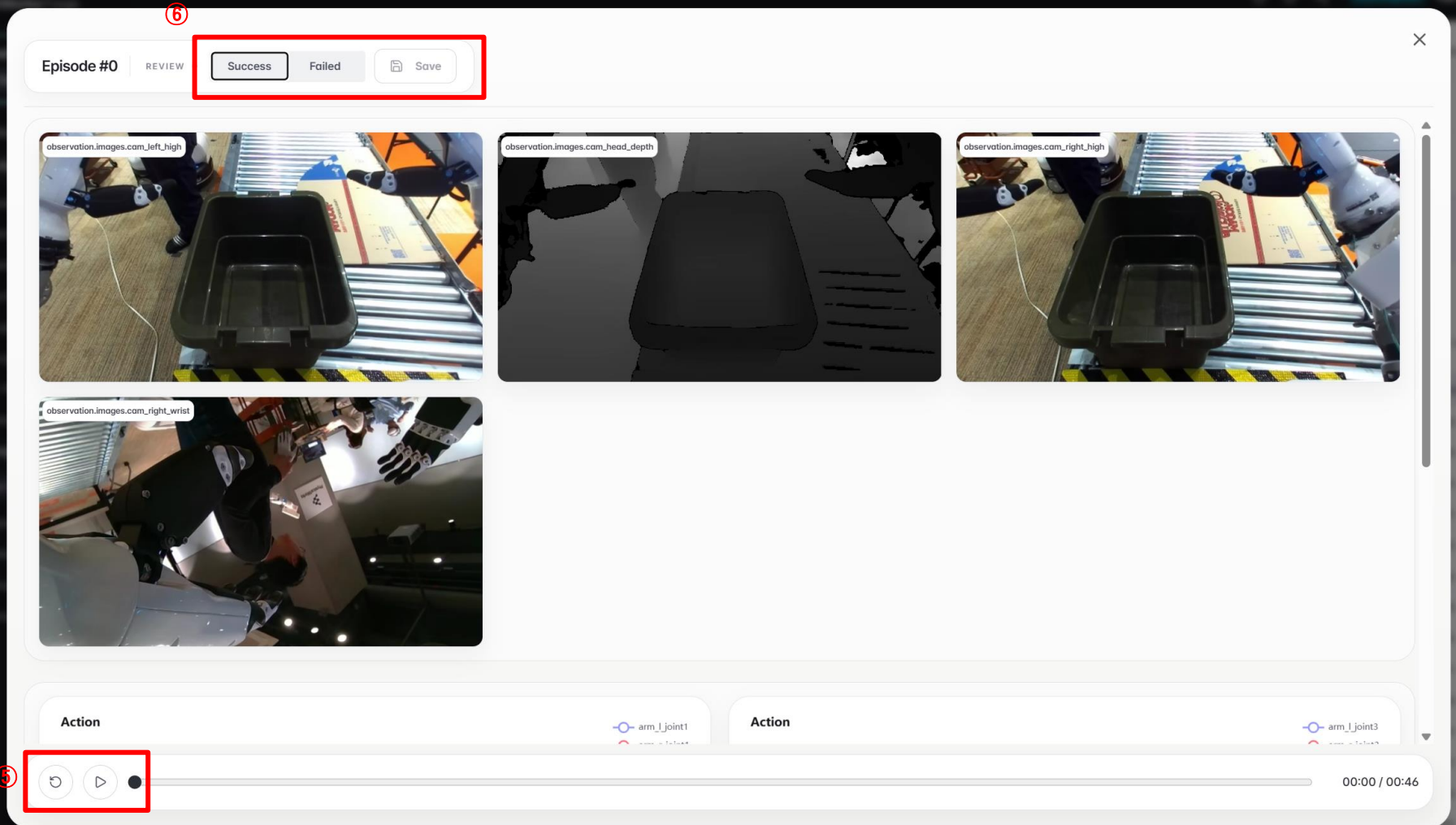
Index	Video	Content	Status	Duration
0		pick up cube.	-	46s
1		pick up cube.	-	49s
2		pick up cube.	-	43s
3		pick up cube.	-	47s
4		pick up cube.	-	42s
5		pick up cube.	-	36s
6		pick up cube.	-	48s
7		pick up cube.	-	55s
8		pick up cube.	-	51s
9		pick up cube.	-	48s
10		pick up cube.	-	32s
11		pick up cube.	-	41s
12		pick up cube.	-	36s
13		pick up cube.	-	36s
14		pick up cube.	-	37s

4. Data Manager (3/4)

저장소에 등록되어 있는 데이터셋 목록을 조회하는 기능

5) Play 버튼 선택 시, 영상과 궤적이 화면이 재생

6) 영상과 궤적 확인 후, Success/Fail 여부를 선택 후, Save 버튼 선택



4. Data Manager (4/4)

저장소에 등록되어 있는 데이터셋 목록을 조회하는 기능

5) Validation Result 탭 선택

6) 최종 결과 확인 > Save Dataset 선택 시, Fail Dataset은 삭제되고 Success 데이터만 저장

The screenshot displays the PhysicalWorks Forge Data Manager interface. On the left sidebar, the 'Vega Logistics Center Project' is selected, with 'Humanoid' and 'Dexmate Vega' sub-projects listed. The 'Data Foundry' section is active, showing options like 'Data Upload', 'Data Collector', 'Data Generator' (with sub-options: WFM Synthetic Data, WFM Augmentation, Mimic Augmentation, Video to Motion), 'Data Curator', 'Data Manager', 'Model Institute', 'Model Trainer', 'Model Evaluator', 'Model Simulation', and 'Model Manager'. The main panel shows the 'G1 exhibition data v2.1' dataset. The 'Validation Result' tab is selected and highlighted with a red box and a circled '7'. The 'Validation Details' section shows: Records (70), Success Records (0 with a green checkmark), Failed Records (0), and Not Reviewed (70). The 'Save Dataset' button is highlighted with a red box and a circled '8'. The 'Validation Records' table is currently empty, displaying 'No data found.' The top right of the interface shows the user 'Data Engineer' and a '개발자' dropdown menu.

PhysicalWorks Forge

Vega Logistics Center Project >
Humanoid Dexmate Vega

Dashboard

Data Foundry

- Data Upload
- Data Collector
- Data Generator
 - WFM Synthetic Data
 - WFM Augmentation
 - Mimic Augmentation
 - Video to Motion
- Data Curator
- Data Manager
- Model Institute
 - Model Trainer
 - Model Evaluator
 - Model Simulation
 - Model Manager

G1 exhibition data v2.1 G1 exhibition data v2.1

Edit Download

Overview Pipeline Episode Data Validation Result

Validation Details

Records 70

Success Records 0 ✓

Failed Records 0

Not Reviewed 70

Auto Validation Save Dataset

Validation Records 1.0.0

Showing 0 of 0 episodes

Index	Duration	Video	Status	Issue
No data found.				

5. Model Trainer (1/2)

저장소에 등록되어 있는 데이터셋 목록을 조회하는 기능

1) Model Trainer 메뉴 선택

2) 기본 정보 입력 > 학습 할 Foundation Model 선택 > Dataset 선택

PhysicalWorks Forge

Vega Logistics Center Project >
Humanoid Dexmate Vega

Dashboard
Data Foundry
Data Upload
Data Collector
Data Generator
WFM Synthetic Data
WFM Augmentation
Mimic Augmentation
Video to Motion
Data Curator
Data Manager
Model Institute
Model Trainer
Model Evaluator
Model Simulation
Model Manager

Model Trainer

Configure, launch, and monitor distributed training runs with real-time metrics.

Step 1: Basic Settings

Model Name*
Enter training name

Description
Enter description (optional)

GPU type

☐ H200 8 Total 3 Running 5 Idle
80GB HBM3e 4.8 TB/s 1,979 TFLOPS FP8
GPUs 1 2 3 4 5 6 7 8

☐ H100 8 Total 3 Running 5 Idle
80GB HBM3 3.35 TB/s 1,979 TFLOPS FP8
GPUs 1 2 3 4 5 6 7 8

☐ A100 8 Total 2 Running 6 Idle
80GB HBM2e 2.0 TB/s 624 TFLOPS FP16
GPUs 1 2 3 4 5 6 7 8

Step 2: Select Foundation Model

Robot Foundation Model*
Select a foundation model Select option

Step 3: Choose Training Dataset

Training Dataset*
Select a dataset from Data Hub Select option

Step 4: Configure Training Parameters

Start Training

5. Model Trainer (2/2)

저장소에 등록되어 있는 데이터셋 목록을 조회하는 기능

3) 직접 Configuration 정보 입력 또는 Load Preset 버튼 선택

4) Load Preset 버튼 선택 시, Preset 선택 가능한 팝업 창이 열림 > Load Preset 선택 시, 해당 값이 자동으로 입력 됨

5) Start Training 버튼 선택 시, 학습이 실행 됨

The screenshot displays the PhysicalWorks Forge Model Trainer interface. The left sidebar shows the project hierarchy: Vega Logistics Center Project > Humanoid > Dexmate Vega. The main panel shows the Model Trainer configuration steps: Step 1 Basic Settings, Step 2 Select Foundation Model, Step 3 Choose Training Dataset, and Step 4 Configure Training Parameters. A 'Load Preset' dialog is open, showing the 'TwinVLA 기본 하이퍼파라미터' preset with the following values: num_gpus: 1, max_steps: 50,000, action_len: 50, batch_size: 4, save_steps: 2,000, and learning_rate: 0. The dialog has 'Cancel' and 'Load Preset' buttons. The 'Load Preset' button is highlighted with a red box and a circled number 7. In the background, the 'Configure Training Parameters' section shows fields for Resume (False), Batch Size (Enter batch size), Max Steps (Enter max steps), and Save Steps (Enter save steps). The '+ Add Preset' and 'Load Preset' buttons are highlighted with a red box and a circled number 6. At the bottom right, the 'Start Training' button is highlighted with a red box and a circled number 8.

PhysicalWorks Forge

Vega Logistics Center Project > Humanoid > Dexmate Vega

Dashboard

Data Foundry

Data Upload

Data Collector

Data Generator

- WFM Synthetic Data
- WFM Augmentation
- Mimic Augmentation
- Video to Motion

Data Curator

Data Manager

Model Institute

- Model Trainer
- Model Evaluator
- Model Simulation
- Model Manager

Model Trainer

Configure, launch, and monitor distributed training

Step 1 Basic Settings

Step 2 Select Foundation Model

Step 3 Choose Training Dataset

Step 4 Configure Training Parameters

Robot Foundation Model

π0.5 (v1.0)

Training Dataset

Select a dataset

Load Preset

TwinVLA 기본 하이퍼파라미터

num_gpus:	1
max_steps:	50,000
action_len:	50
batch_size:	4
save_steps:	2,000
learning_rate:	0

Cancel Load Preset

Configure Training Parameters

Resume

False

Batch Size

Enter batch size

Max Steps

Enter max steps

Save Steps

Enter save steps

+ Add Preset Load Preset

Start Training

6. Model Simulation

모델 추론 결과를 Simulation 환경을 통해 검증해 볼 수 있는 기능

1) Model Simulation 메뉴 버튼 선택

2) Connect 버튼 선택 > Simulation 서버 접속 및 화면 표시 > Run 버튼 선택 > Simulation 동작 수행

The screenshot displays the PhysicalWorks Forge interface for Model Simulation. The sidebar on the left contains the following navigation items: Dashboard, Data Foundry, Data Upload, Data Collector, Data Generator (with sub-items: WFM Synthetic Data, WFM Augmentation, Mimic Augmentation, Video to Motion), Data Curator, Data Manager, Model Institute (highlighted with a red box and a circled '1'), Model Trainer, Model Evaluator, Model Simulation (highlighted with a red box and a circled '1'), and Model Manager. The main area is titled 'Model Simulation' and includes a subtitle: 'Connect 버튼으로 Omniverse WebRTC 스트림에 연결합니다.' The central 'SIMULATION VIEWER' is titled 'Warehouse Pick' and shows 'Bin picking and object placement workflow for validation.' Below the viewer are buttons for 'Server 10.81.15.238', 'Latency - ms', and 'FPS -'. The right panel, 'Simulator Controller', lists three simulation types: 'Warehouse Pick' (Bin picking and object placement workflow for validation), 'Assembly Line' (Repeated arm motion and task sequencing in a constrained cell), and 'Inspection Route' (Mobile manipulation route with checkpoints and camera review). Below these are buttons for 'Connect' (highlighted with a red box and a circled '2'), 'Disconnect', 'Run' (highlighted with a red box and a circled '3'), 'Pause', and 'Reset'. The bottom section shows a log of events: '16:30:12 Simulation workspace initialized.' and '16:30:28 Viewer ready. Waiting for a scene selection.'

7. Model Manager (1/2)

저장소에 등록되어 있는 Model 목록을 조회하는 기능

1) Model Trainer 메뉴 선택

2) Model 목록을 확인하고, 상세 정보 또는 다운로드를 원하는 Model 항목을 선택

PhysicalWorks Forge

Vega Logistics Center Project >
Humanoid Dexmate Vega

Dashboard

Data Foundry

Data Upload

Data Collector

Data Generator

- WFM Synthetic Data
- WFM Augmentation
- Mimic Augmentation
- Video to Motion

Data Curator

Data Manager

Model Institute

- Model Trainer
- Model Evaluator
- Model Simulation
- Model Manager

Model Manager

Browse and manage all RFM in one place.

Q Search

All

post-trained v2.5

Cosmos Transfer 2.5 Base

Parameter Size 2B Size —
Last Updated 2026-05-13 17:00

pre-trained v1.0

π0.5

Parameter Size 3B Size 26.9 MB
Last Updated 2026-05-14 17:06

pre-trained v1.0

TwinVLA

Parameter Size 1B Size 2.0 GB
Last Updated 2026-04-20 15:10

pre-trained v1.0

ACT

Parameter Size 80M Size 197.2 MB
Last Updated 2026-04-14 15:05

pre-trained v1.0.0

TwinVLA

Parameter Size 1B Size 82.2 GB
Last Updated 2026-04-01 03:02

pre-trained v2.5

Cosmos-Predict2.5-2B/Pre-train...

Parameter Size 2B Size 3.8 GB
Last Updated 2026-03-17 10:38

pre-trained v1.0

GR00T N1.5

Parameter Size 3B Size 1.9 GB
Last Updated 2026-03-17 10:38

post-trained v2.5

Cosmos-Predict2.5-14B/Post-tra...

Parameter Size 14B Size 26.8 GB
Last Updated 2026-03-17 10:38

pre-trained v2.5

Cosmos-Predict2.5-14B/Pre-trai...

Parameter Size 14B Size 26.8 GB
Last Updated 2026-03-17 10:38

post-trained v2.5

Cosmos-Predict2.5-2B/Post-trai...

Parameter Size 2B Size 3.8 GB
Last Updated 2026-03-17 10:38

7. Model Manager (2/2)

저장소에 등록되어 있는 Model 목록을 조회하는 기능

3) Edit 버튼 선택 시, Model 정보 수정이 가능 함

4) Download 버튼 선택 시, 지정된 Local로 해당 Model 파일이 전송됨

The screenshot displays the PhysicalWorks Forge Model Manager interface. On the left is a dark sidebar with navigation options: Dashboard, Data Foundry, Data Upload, Data Collector, Data Generator (with sub-items: WFM Synthetic Data, WFM Augmentation, Mimic Augmentation, Video to Motion), Data Curator, Data Manager, Model Institute (highlighted), Model Trainer, Model Evaluator, Model Simulation, and Model Manager. The main content area shows the details for the 'Cosmos-Predict2.5-14B/Post-trained' model. At the top right, there are icons for notifications, search, and user profile, along with the text 'Data Engineer 개발자'. Below these are two buttons: 'Edit' (with a pencil icon) and 'Download' (with a download icon), both highlighted with red boxes and numbered 1 and 2 respectively. The model details are presented in three cards: 'Size' (26.8 GB), 'Foundation Model' (Cosmos-Predict2.5), and 'Training Dataset'. Below these is a 'Model Information' table.

Model Information			
Model Name	Cosmos-Predict2.5-14B/Post-trained	Description	Given a text description, an image as the first frame, and/or a video, predict the ...
Version	v2.5	Parameter Size	14B
Created Date	2026-03-17	Last Modified	2026-03-17
Owner	system		

감사합니다.